

BLOCK 3

Reasoning Strategies & RL

Part 1: CoT + Self-Consistency · Part 2: Why models do what they do

Block 3 — 3 hours

The Bar Exam result

GPT-3.5 failed. GPT-4 top 10%. Same family. One phrase added.

Zero-shot CoT

'Think step by step.' Free. Immediate effect on multi-step tasks.

Few-shot CoT

EVIDENCE / ANALYSIS / CONCLUSION / CONFIDENCE format. Teach your domain reasoning.

Self-Consistency

k=3 reasoning paths. Majority vote. +5–10 pp faithfulness.

RLHF, RLAI, RLVR

Why your system prompt works. Why function calling is reliable.

Reward hacking

Your performance measure IS a reward function. Audit and fix it.

RAGAS + Deliverable 3

Final table. Block 1 baseline vs Block 3 results.

01

CoT & Self-Consistency

From guessing to measurable reasoning

The Bar Exam — what changed

GPT-4 scored in the top 10% of the US Bar Exam. GPT-3.5 failed it.
Same architectural family. Same training data. One phrase added to the prompt.

What changed	Why it works
The prompting strategy	Intermediate reasoning tokens constrain the final answer token.
Not the model	Same parameters, same training. Better reasoning quality is available now.
Not magic	CoT reduces the error rate — it does not guarantee correctness. The model can still make mistakes.
The practical implication	Change the prompt, not the model. Better reasoning = free, immediate.

When to use CoT: multi-step tasks, multi-factor analysis, synthesis of conflicting sources.

When NOT to use: simple lookups, tool selection ($T=0$), creative writing.

Few-Shot CoT — The EVIDENCE / ANALYSIS / CONCLUSION / CONFIDENCE format

Zero-shot CoT produces generic reasoning. Few-shot CoT teaches the exact reasoning format for your domain.

EVIDENCE

List each retrieved source with attribution:

- UNHCR 2023: 21.5M displaced/year
- World Bank 2024: Mekong Delta -40%

Purpose: makes reasoning auditable.
Your critic agent can verify each entry independently.

ANALYSIS

- Step 1: [first reasoning step]
- Step 2: [second reasoning step]
- Step 3: [reconcile conflicts]

Purpose: forces decomposition before the conclusion.

Prevents jumping to an answer before processing all elements.

CONCLUSION + CONFIDENCE

CONCLUSION: [your answer]

CONFIDENCE: HIGH / MEDIUM / LOW

→ one-sentence justification

LOW confidence = trigger another search pass.

HIGH confidence on a single source = suspicious.

Non-negotiable: an agent that expresses uncertainty is more reliable than one that confidently hallucinates.

Self-Consistency — k paths, majority vote

Generate k independent reasoning chains · Take the majority answer · Cost: k× the tokens

	Single path	Self-Consistency k=3
Mechanism	One reasoning path at T=0.1	Three paths at T=0.8 (high diversity), majority vote
Why it helps	No check — the first error propagates	A wrong answer requires a specific bias — unlikely to emerge in majority
RAGAS gain	Baseline	↑ faithfulness +5–10 pp on complex analytical tasks
Cost	1× tokens	k× tokens — reserve for the final synthesis step only
When to use	Simple retrieval, tool selection	High-stakes final synthesis — never at every step

02

Why Models Do What They Do

RLHF · RLAIIF · RLVR · reward hacking

The three RL variants and what they produced

Variant	Reward signal	What emerged	Why you care
RLHF	Human preference comparisons	Instruction-following LLMs (GPT, Claude)	Your system prompt works because of RLHF
RLAIF	AI evaluator applying principles	Constitutional AI — scalable alignment	Your critic agent IS RLAIF at a smaller scale
RLVR	Verifiable correctness (maths/code)	Reasoning models (DeepSeek-R1, o1)	Extended thinking on hard analytical tasks
Tool-use RL	Correct tool call + correct answer	Reliable structured JSON function calling	The entire agent loop depends on this
Reward hacking	(failure mode — not a variant)	\$6.2M AWS incident	Your performance measure IS a reward function

RLVR — Why reasoning models reason better

What RLVR trains on

Tasks with VERIFIABLE answers:

- Mathematical proofs
- Code that passes unit tests
- Logical deduction

Reward signal: binary.

Correct = 1. Incorrect = 0.

No human labellers needed.

Fully automated training signal.

The model learns: longer, more careful reasoning → more correct answers.

What emerged

Models that generate extended thinking chains:

- Self-correct mid-reasoning
- Backtrack when they detect an error

DeepSeek-R1:

- Free on Groq
- Open weights — runs locally
- Comparable to o1 on maths/logic
- \$2.19/M tokens vs \$15 for GPT-4o

Groq model: deepseek-r1-distill-llama-70b

When to use in your agent

Use DeepSeek-R1 for:

- Complex multi-factor synthesis
- Reconciling conflicting sources
- High-stakes analytical tasks

Keep Llama-3.3-70b for:

- Tool selection (fast, T=0)
- Simple retrieval
- High-frequency calls

Lab: compare both on your 5 questions.
Note which type benefits most.

Reward Hacking — The lesson for your agent

**Your performance measure in the system prompt IS a reward function.
Any incomplete measure can be 'hacked'.**

Hackable measure

'Complete the research task efficiently.'

Optimal hacking strategy:
Empty response immediately.

- 0 tokens → minimal cost ✓
- Zero latency → maximum speed ✓
- Task 'completed' ✓

Reward: maximised.

Usefulness: zero.

Robust measure

- ACCURACY: every claim traces to a URL
- COMPLETENESS: address all aspects
- CONFIDENCE: express uncertainty

Failure modes to exclude:

- ✗ Empty answer when sources exist
- ✗ Citing a URL not in context
- ✗ HIGH confidence with one source

The attacker can no longer exploit a favourable interpretation.

Immediate action: re-read your system prompt. What is the most efficient strategy to hack it while being useless? Add constraints that exclude that strategy.

Deliverable 3 — Final RAGAS table

Re-run RAGAS one last time. Compare Block 1 baseline vs Block 3 final. This table goes in REPORT.md.

Metric	Block 1 Baseline	After Block 3	Primary technique
context_recall	---	---	Block 1: hybrid search + parent-child
context_precision	---	---	Block 1: cross-encoder reranking
faithfulness	---	---	Block 3: Self-Consistency + few-shot CoT
answer_relevancy	---	---	Block 3: EVIDENCE/ANALYSIS/CONCLUSION format

Deliverable 3 — Final RAGAS table

✓ Deliverable 3 — show the instructor before leaving

- ☐ Few-shot CoT with EVIDENCE/ANALYSIS/CONCLUSION/CONFIDENCE in your synthesis prompt
- ☐ Self-Consistency k=3 implemented and tested
- ☐ RAGAS table: Block 1 baseline vs Block 3 results